Observational evidence for more than a meter of 21st century sea-level rise

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Abstract

Every centimeter of sea-level rise exposes roughly one million additional people to coastal flooding, yet the projections guiding adaptation planning rely on complex ice-sheet models that diverge below observed trends within years. We construct a simple, physical, data-driven baseline forecast by calibrating individual sea-level contributors with independent observations and validating against observed global mean sea level (GMSL). Our forecasts show that GMSL will likely rise 1.0–1.3 m (medians across policy-relevant pathways) by 2100, exceeding IPCC estimates by 70–79% for the same warming, with a 71% probability of exceeding one meter averaged across all policy-relevant scenarios. West Antarctic glacier dynamics accounts for 92% of forecast uncertainty, underscoring the importance of both rapid reductions in CO_2 emissions and planning for rapid loss of ice in West Antarctica.

Rates of global mean sea-level rise have doubled twice in the past 50 years [1, 2] (Fig. 1). The first doubling took about 16 years and 0.34 °C of warming [3]. The second took about 21 years and 0.41 °C of warming (Fig. 1). The rate could double again by roughly 2080 if the current observed sea-level acceleration continues, and by around 2040 if it takes no more warming than the previous two doublings. The 40-year gap between these two simple, data-driven predictions reflects the uncertainty that coastal adaptation must absorb.

Every centimeter of sea-level rise exposes roughly one million additional people to annual coastal flooding [7], approximately 90% of whom live in low- and middle-income countries [8]. Without additional adaptation, annual flood damages are projected to reach US$14–18 trillion per year for 0.9–1.1 m of rise [9]. The cost of adaptation is projected at US$40–170 billion per year by mid-century [10], yet developing countries currently receive US$21 billion per year in total adaptation finance across all sectors, not just coastal [11], raising serious questions about the feasibility of coastal adaptation at the scale required and highlighting the value of reliable sea-level forecasts.

Most coastal adaptation planning relies on projections from the IPCC Sixth Assessment Report (AR6), which estimates 0.4–0.7 m of rise by 2100 relative to the 1995–2014 baseline [12, 13]. These projections depend largely on process-based ice sheet models that diverge from satellite observations within years of initialization [14, 15]. As a result, observed GMSL trends (0.66 m median at 2100, 0.41–0.91 m 90% CI) already exceed AR6 medium-confidence projections under all policy-relevant emission scenarios [2, 16]. AR6 itself documents this in a single figure but without comment: extrapolating the satellite-era rate and acceleration—which reflect the current warming trajectory that could lead to 3 $\circ$C (SSP2-4.5)—produces sea-level rise that matches or slightly exceeds the process-based projections under very high emissions scenario (SSP5-8.5), which is now regarded as generating an implausible level of warming (AR6 Figure 9.27; [12, 17]). So while the data show that rates of sea level rise double with approximately 0.4 °C of warming (Fig. 1b), IPCC models need an order of magnitude more warming to match current rates of sea-level rise.

Disagreement between observations and ice sheet models arise from known structural biases, including the absence of modeled iceberg calving [15] and an assumed ice viscosity that is less sensitive to stress than satellite and laboratory observations indicate [18, 19]. The latter produces a 21–35% underestimate of projected 21st century ice loss [20, 21], while the absence of calving produces an unquantified error that is likely leading order based on the observed sensitivity of glacier flow to some major calving events [22]. More broadly, decades of general forecasting research shows that models more complex than their calibration data can constrain produce worse out-of-sample predictions than simpler models [23, 24]. Current ice sheet models have millions of degrees of freedom representing spatially distributed parameters and uncertain parameterizations. The observations used to initialize these models inform only a small fraction of these degrees of freedom, with the remainder determined by prior assumptions [25, 26].

Data-driven forecasting

We construct a data-driven, Bayesian forecast that inverts this ratio by focusing on different sea-level contributors, each with minimal physical models constrained by independent observational records spanning decades. Observed GMSL and its sensitivity to observed global mean surface temperature (GMST) are diagnostics, and we show that the sum of our model components agrees within 6% of 2000–2025 observed GMSL in hindcasting tests (Supplementary Material). Unless otherwise noted, all sea-level values in the main text are 21st-century contributions referenced to the year 2000, while values in the abstract are relative to pre-industrial for consistency with existing projections and literature. Throughout, we label emission scenarios by their approximate end-of-century global mean surface temperature: 2 °C (SSP1-2.6), 3 °C (SSP2-4.5), 4 °C (SSP3-7.0), and 5 °C (SSP5-8.5).

Our framework builds on two established lines of data-driven forecasting. The simplest is a fit to the rate and acceleration from GMSL observations, which requires no physical model nor assumptions. The satellite altimetry record provides GMSL measurements from 1993–2025 that show a constant acceleration of 0.08 $\mathrm{mm\,yr^{-2}}$ (0.02–0.14 $\mathrm{mm\,yr^{-2}}$) and rate 4.5 $\mathrm{mm\,yr^{-1}}$ (3.5–5.5 $\mathrm{mm\,yr^{-1}}$) in 2025 [2, 16]. Extrapolating these values to 2100 yields 66 cm (41–91 cm) of 21st-century rise (Fig. 3a). Assuming the naive forecast (extrapolating observed trends) has predictive skill over at least half the calibration window, the extrapolation of the 32-year satellite record provides predictive skill over the next 15-20 years. Beyond that horizon, the quadratic is less reliable but still establishes a scenario-independent reference (Fig. 3a).

At the next level of model complexity are the semi-empirical models that are based on the sensitivity of GMSL rise rate to changes in GMST (Fig. 1b). Under current warming trends, Rahmstorf (2007) [5] predicts 0.67 m (0.63–0.71 m) of 21st-century rise, matching the extrapolation of modern observed rates within error [2]. An updated semi-empirical model [6], which adds a rate-of-warming term, predicts 0.96 m (0.80–1.12 m), in close agreement with the results of this study. Although physically motivated by the fit to warming sensitivities rather than just sea-level trends, semi-empirical models were assessed as *low confidence* by IPCC AR5 because they do not resolve individual contributing processes and because extrapolation beyond the calibration period assumes a stationary temperature–sea-level relationship [27]. This reasoning is carried forward by AR6 but contradicts the decisions of AR5 and AR6 to favor process models that omit key physics such as calving, use untested parameterizations such as ocean-driven melting that have leading-order effect on projections, are initialized with static fields of non-stationary data defined with much narrower calibration windows, and were never tested against observed GMSL, the very quantity they attempt to predict (cf. AR6 Figure 9.27 in [12]). In short, IPCC AR5 and AR6 set aside simple, data-driven models that predict observations in favor of complex, process models that fail to reproduce observed trends [14].

Component model calibration

We decompose global mean sea level into separate components: ocean thermal expansion (thermosteric), glaciers (not in the ice sheets), Greenland Ice Sheet (GrIS) surface mass balance (SMB), GrIS ice discharge, West Antarctic Ice Sheet (WAIS) discharge, East Antarctic Ice Sheet (EAIS) SMB, Antarctic Peninsula total ice loss, and terrestrial water storage (TWS). This component-summation approach builds on the probabilistic framework of Kopp et al. [28], the observation-calibrated semi-empirical decomposition of Mengel et al. [29], and the jointly calibrated Bayesian architecture of Wong et al. [30], while extending these approaches with rate-space blending against satellite-era observations and a structured deep-uncertainty treatment of WAIS dynamics. The sensitivities of glaciers, GrIS discharge, and Antarctic Peninsula to warming are calibrated with independent observational datasets spanning 24–71 years (Table S1). Surface mass balance for GrIS and EAIS is the one component where we adopt published model sensitivities rather than calibrating directly, because the relevant processes (melt-albedo feedback, refreezing) require spatially resolved energy balance modeling and lack sufficient data coverage for our approach [31, 32].

The thermosteric component—the largest single contributor to observed sea-level rise but a small contributor to forecast uncertainty—is modeled with a two-layer energy balance [35] fitted jointly to NOAA in situ thermosteric sea level measurements at 0–700 m (1957–2025) and 0–2000 m (2005–2025) [36] (Fig. 2a). In our model, the upper ocean responds to GMST warming through a thermal state variable S_u with relaxation time τ_u. The deep ocean relaxes toward S_u on a longer timescale $\tau_d = 150\,\mathrm{yr}$ [35]. To account for ocean expansion below-2000 m, we add a rate correction of $0.07\,\mathrm{mm\,yr^{-1}}$ [37]. We apply a 14% upward correction to NOAA observations to account for the documented low bias arising from the interpolation of sparse hydrographic profiles, primarily in the Southern Ocean [38]. We carry a 5% residual structural uncertainty in the thermosteric fits and forecast.

The Greenland Ice Sheet is currently the largest land-ice contributor [1], but not the largest source of land-ice uncertainty. We treat GrIS SMB and ice discharge separately. SMB observations from a regional climate model (RCM), RACMO [33, 39], are used directly for the hindcast. The forecasts scale with temperature using sensitivities from regional climate model intercomparisons [31, 32]. Our choice of RCM does not materially affect the hindcast (4% difference; Supplementary Material). RACMO sits at the conservative end of the RCM range: all models systematically underestimate ablation-zone melt relative to in situ observations [31], and RACMO's bare-ice albedo remains too high despite partial MODIS assimilation [40, 41]. Under strong warming, the inter-RCM spread becomes a factor of two in projected SMB [42]. This structural uncertainty is captured in our forecast priors.

GrIS discharge is modeled as the cumulative response to time-delayed subsurface ocean warming, with a sensitivity γ, a delay δ between ocean warming and ice sheet response, and a secular drift r_0 from post–Little Ice Age adjustment (Supplementary Material). Cross-correlation of EN4 subsurface ocean temperature [43] against the Mouginot et al. [33] discharge record and Bayesian Information Criterion (BIC) model selection yields $\delta = 5$ yr (Fig. 2b). The fitted sensitivity $\gamma = 0.37\ \mathrm{mm\,yr^{-1}\,°C^{-1}}$ agrees with physical scaling estimates from ocean and glacier dynamics ($0.36\ \mathrm{mm\,yr^{-1}\,°C^{-1}}$). An independently derived discharge record (1986–2023) [44] and NOAA Arctic Report Card estimates [45], withheld from calibration, fall within the 90% credible interval.

Mountain glacier mass loss is modeled as a rate–temperature relationship calibrated against the GlaMBIE global consensus [2000–2023; 46]. The data only support a linear sensitivity (Supplementary Material), which we find to be $b_{\mathrm{gl}} = 0.66\ \mathrm{mm\,yr^{-1}\,°C^{-1}}$ (Fig. 2c). This is approximately five times the value embedded in IPCC AR6 median projections ($\sim$0.13 $\mathrm{mm\,yr^{-1}\,°C^{-1}}$) and 1.3–1.7× the effective sensitivity of the PyGEM process model [47], but is overall a small contributor to sea-level rise (Supplementary Materials).

We compare the sum of these independently calibrated components to observations of GMSL from satellite altimetry as a diagnostic (Fig. 2d,e). The diagnostic is possible because each component is calibrated against its own observational record; total GMSL is not used as a calibration target. The component sum closes 90–94% of the observed GMSL rate over 1993–2025, with a variance-weighted attribution assigning the residual mainly to the thermosteric (61%) and terrestrial water storage (21%) components, both within their stated uncertainties (Supplementary Material). All components are within 1.4 standard deviations of their independent estimates after variance-weighted redistribution of the residual.

West Antarctic Ice Sheet. WAIS is the dominant source of forecast uncertainty (92% of within-scenario variance at 2100; Figs. 2f and 3b). The ice sheet contains $\sim$3.3 m of sea level equivalent on retrograde beds susceptible to marine ice sheet instability (MISI; [48]), and ocean thermal forcing sufficient to drive retreat is committed regardless of emissions pathway [49, 50]. We therefore make our WAIS projections independent of the warming trajectory, meaning the range of 21st-century outcomes is controlled by ice dynamics, not emissions.

No physical process models have been proven to produce reliable projections of WAIS (Fig. 2f). To provide a quantified, representative range of uncertainties, we construct a scenario-mixture framework with three outcomes that are grounded in the scientific literature (Supplementary Material): (S1) 'stable WAIS' with little-to-no internal dynamical feedbacks (MISI), where discharge continues in the range of present-day rates (25–85 mm by 2100) is given a 10% probability; (S2) MISI-driven acceleration with grounding-line retreat on retrograde bedrock (150–1000 mm) has an 80% probability; and (S3) MISI combined with marine ice cliff instability (600–1400 mm) is assigned a 10% probability. The S2 lower bound is anchored to transient-calibrated ice sheet model projections [51] and the observed 33 km grounding-line retreat at Pine Island Glacier since 1992 [52]. Within each scenario, the 2100 endpoint is drawn from a skew-normal distribution in log-space with physically motivated asymmetry [53]. A multiplicative rheology correction (R with median $R = 1.31$ at the observed $n = 4.1$ [18]) is applied to all scenarios, reflecting the 21–35% underestimation of ice loss when models assume $n = 3$ [20, 21]. The resulting mixture (median 0.41 m [0.07, 1.56 m] at 2100) is wider than the structured expert judgment of Bamber et al. [54] (which predates the rheology correction) but in broad agreement. Our choices of scenario weighting, with 10-80-10% weighting from S1 to S3, have a small quantitative impact on our forecast results in the high-end tails of the distribution, but do not qualitatively impact the findings of this study (Supplementary Materials).

Forecasts

We focus on warming trajectories of 2 °C (SSP1-2.6) through 4 °C (SSP3-7.0), the pathways most consistent with current policies and stated pledges (Fig. 3). Under 2 °C target (aggressive emissions cuts and including mid-century overshoot), the blended forecast projects 0.78 m of 21st-century sea-level rise by 2100 [5th–95th percentile: 0.44–1.92 m], 79% above the IPCC AR6 medium-confidence estimate of 0.44 m. Under 3 °C warming (intermediate emissions), the median is 0.95 m [0.61–2.08 m], 70% above IPCC's 0.56 m. For the 4 °C trajectory, the median reaches 1.15 m [0.80–2.29 m]. For reference, 5 °C warming trajectory (SSP5-8.5) yields a median of 1.34 m [0.98–2.48 m]. All sea-level values given relative to year 2000.

Relative to pre-industrial GMSL (adding ~0.19 m of observed rise through 2000), these forecasts correspond to medians of 0.97–1.34 m of global mean sea level rise across the policy-relevant pathways. The probability of exceeding 1 m relative to pre-industrial by 2100 is 46% under 2 °C warming, 74% under 3 °C, and 95% under 4 °C warming trajectory. Averaged across these pathways, the probability of exceeding 1 m of sea level rise relative to pre-industrial levels is 71%, and is largely insensitive to the relative weighting of scenarios (Fig. 3).

The within-scenario 90% uncertainty range (1.47 m for 3 °C) is four times larger than the spread across scenario medians (0.37 m from 2–4 °C). A variance decomposition (law of total variance across policy scenarios) attributes 93% of total forecast variance to within-scenario uncertainty—predominantly WAIS—and 7% to between-scenario differences. This reflects the dominance of WAIS uncertainty, which is largely independent of the emission pathway, because ocean thermal forcing sufficient to sustain WAIS retreat is committed regardless of emissions [49]. The between-scenario spread arises from the temperature-sensitive components—primarily glaciers and the Greenland Ice Sheet—whose observationally calibrated sensitivities substantially exceed those in current consensus projections. The discrepancy with IPCC medium-confidence results is accordingly largest under low warming (1.8× for 2 °C) and narrows slightly toward high warming (1.7× for 4 °C). Our medians fall within the IPCC low-confidence ranges, which incorporate structured expert judgment on ice sheet instabilities but are excluded from the projections most commonly used for coastal planning [12, 13]. Our forecasts thus provide quantified probabilities for outcomes that AR6 treats as low-likelihood, high-impact storylines without assigning probabilities.

Implications

We highlight main implications for coastal adaptation planning that follow from our results. First, the central estimates of sea level rise used for infrastructure decisions are too low, likely by a factor of 1.7–1.8, with the largest discrepancy under the moderate-warming scenarios most relevant to current policy. Second, the uncertainty is wider than current assessments acknowledge***not explicitly true; need to polish wording***, with West Antarctic ice dynamics the largest contributor by far. Third, our decomposition reveals a sharp divide between the predictable and unpredictable components of sea-level rise.

The thermodynamic components—ocean thermal expansion, glacier mass loss, Greenland surface mass balance and glacier discharge, and the Antarctic Peninsula—respond to global temperature through well-characterized physical relationships calibrated against decades of observations. Together, these components account for only ∼8% of total forecast uncertainty. Their contributions are predictable in the sense that they are directly attributable to warming in a manner that is constrained by data, reducible by emissions reduction, and distinguishable across warming trajectories. The purple band in Figure 3a shows the forecast distribution if WAIS were to lose mass slowly and predictably: a narrow range of 0.4–0.7 m that is both lower and far more certain than the full forecast. The remaining 92% of forecast uncertainty originates from a single source in the dynamics of glaciers in WAIS, especially Pine Island and Thwaites Glaciers. Their future is not constrained by existing process model results (Fig. 2f) nor is it reducible by emissions reductions on 21st-century timescales [49, 50]. WAIS alone transforms a predictable, moderate-rise future into a costly and uncertain future.

The exceedance curves (Fig. 3b) translate our forecasts into human and economic terms. Under the 3 °C trajectory, the median rise of 0.95 m could expose an additional ∼100 million people to annual flooding [7]. Our forecasts assign a 40% probability of exceeding 1.5 m—a threshold at which flood damages reach US$14–18 trillion/year without additional adaptation [9]. Even under the 2 °C trajectory, the probability of exceeding 1 m relative to pre-industrial is 46%, an outcome for which current adaptation finance—US$21 billion/year across all sectors in developing countries [11]—falls short by at least an order of magnitude relative to projected adaptation costs of US$40–170 billion/year [10]. That the exceedance curves are nearly parallel across warming trajectories (Fig. 3b) underscores that emissions reductions alone cannot eliminate the risk of high sea-level outcomes. Adequate adaptation planning must account for the full range of West Antarctic uncertainty regardless of the emissions pathway achieved.

The data-driven framework presented here is not a replacement for process models, which remain essential for understanding the mechanisms of ice sheet change. Our framework is complementary: an independent, data-driven estimate that makes its assumptions and limitations explicit, provides an observational benchmark against which process models can be tested, and produces conditional distributions that reflect what the data currently support. We consider robust, well-calibrated process models to be a high-value targets for research and development, but the evidence suggests these models are not ready to be used for planning that costs billions of dollars and impacts millions of lives. We emphasize that the disconnect between the the high costs of sea level rise and the large uncertainties in current sea level projections is a consequence of chronic under-investment compounded by the lack of institutional homes able to focus on translating knowledge gained through basic scientific research into forecasts for coastal communities in the way that, for example, the National Weather Service has traditionally done [55]. A specific example is that modeling efforts for IMSIP6, which informed IPCC AR6 projections, was largely unfunded, leading to a dearth of model runs and near-absence of model parameter-sensitivity studies [14]. The world is going to endure high costs from sea-level rise (Fig. 3b), the questions are: how high will those costs rise, and will we decide to invest more heavily now to understand where and how to target funding to reduce the harm to people, ecosystems, and infrastructure in the future, or continue on our current path?

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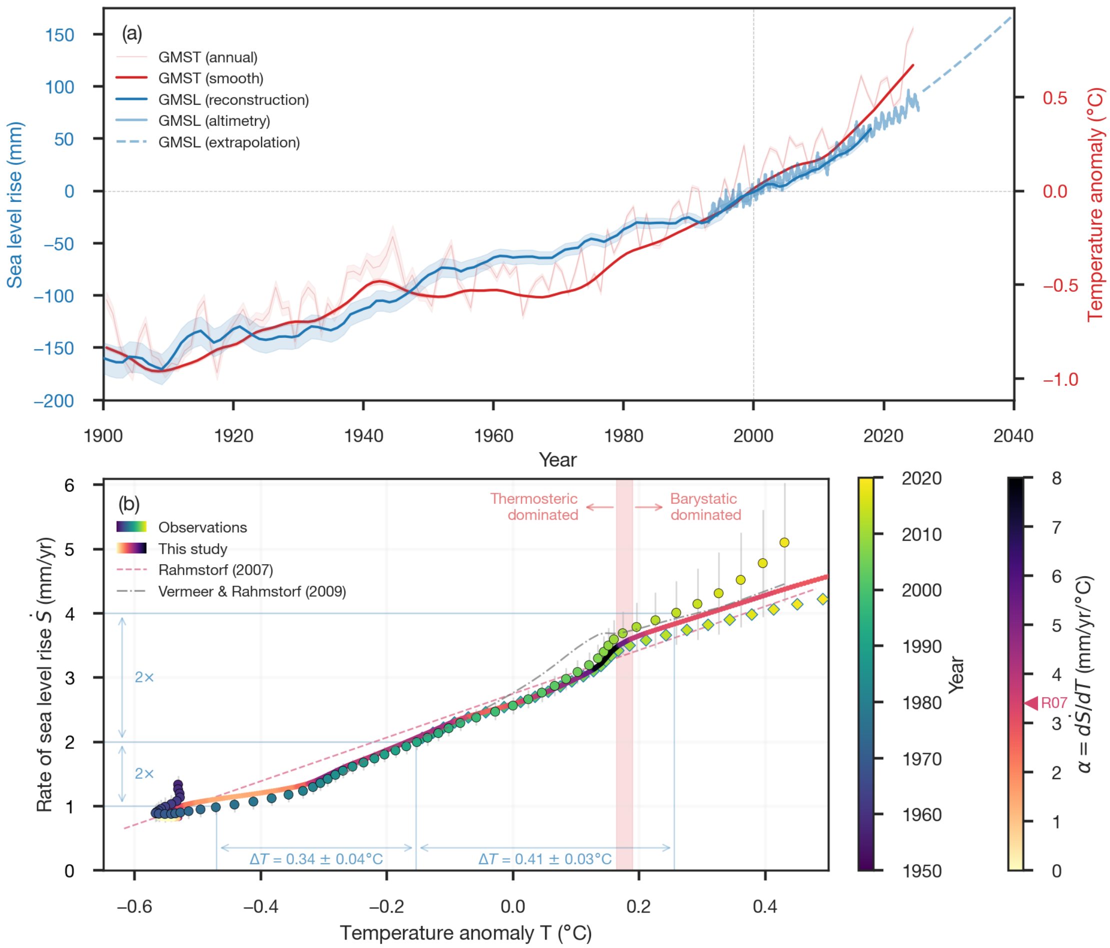

Figure 1: **Observed global mean sea-level rise and its response to global mean surface temperature. (a)** GMSL (blue, 5-year smooth; [1]) and NASA satellite altimetry (light blue; [4]), with the satellite-era quadratic extrapolation (dashed; [2]) and GMST (red, 15-year smooth [3]) on the right axis. **(b)** Rate of change in GMSL ($\dot{S} = dS/dt$) versus GMST anomaly, T. Observed values are shown in circles, colored by year, and the Bayesian rate-and-state semi-empirical model of this study are colored by instantaneous transient sensitivity $\alpha = d\dot{S}/dT$. Other semi-empirical model results are shown as dashed [5] and dash-dot lines [6]. Horizontal and vertical blue lines mark the 1–2 and 2–4 mm yr^{-1} rate doublings that occurred during 1976–1992 (1959–1993, $\pm 1\sigma$) and 1992–2013 (1992–2014, $\pm 1\sigma$), respectively. The corresponding warming increments (ΔT) show that the first doubling required less warming (0.34 ± 0.04 °C) than the second (0.41 ± 0.03 °C). The vertical red band indicates the transition from thermosteric-dominated to barystatic-dominated sea-level rise (ocean mass change due to glaciers, ice sheets, and terrestrial water storage). The sinuous tail in the early observations is largely due to dam impoundment of water rather than a thermodynamic signal [1].

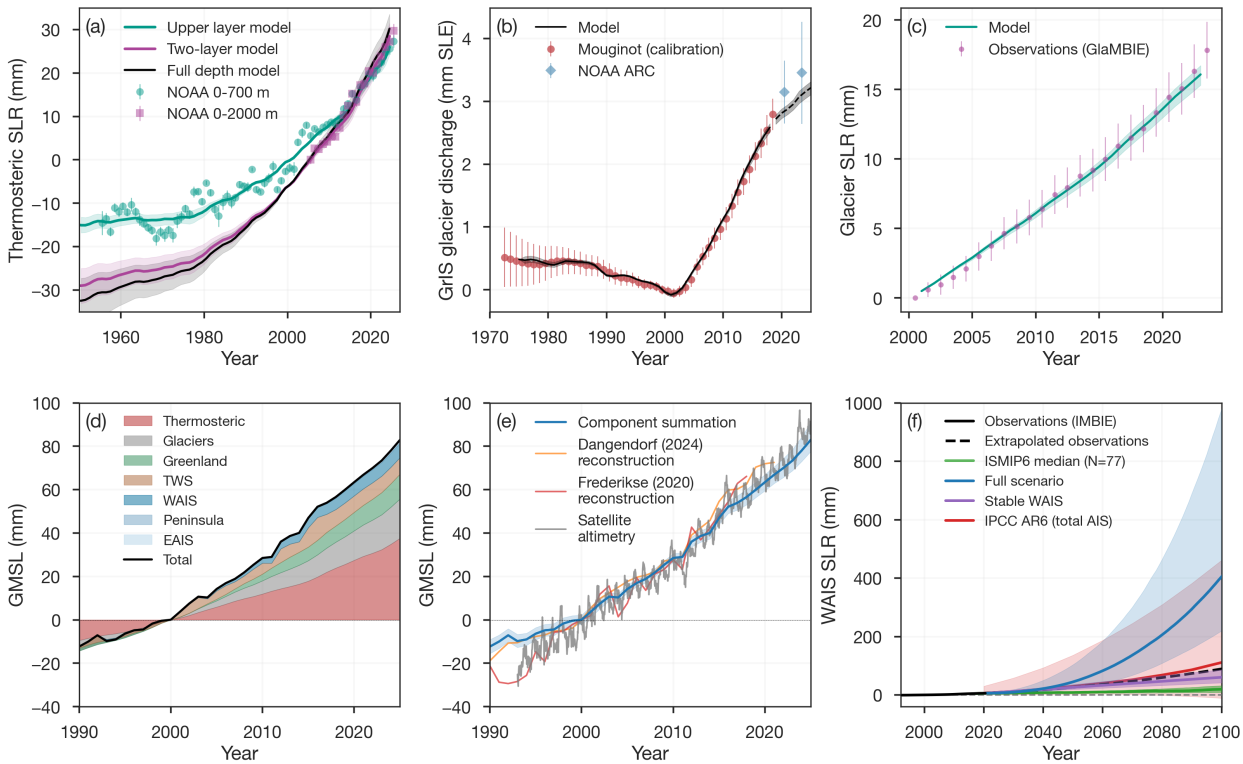

Figure 2: **Model calibration, validation, and WAIS uncertainty. (a)** Two-layer energy balance model fitted jointly to NOAA thermosteric sea level at 0–700 m (1957–2025) and 0–2000 m (2005–2025). **(b)** Greenland ice discharge modeled as a cumulative time-delay response to subsurface ocean warming ($\delta = 5$ yr, $R^2 = 0.995$). Calibration data (Mouginot et al. (2019) [33], filled circles) and withheld validation data (NOAA Arctic Report Card, diamonds) both fall within the 90% credible interval. **(c)** Global glacier contribution fitted to GlaMBIE consensus observations (2000–2023) using a linear temperature-driven model ($R^2 = 0.995$). Grey band shows the 90% credible interval. **(d, e)** Budget closure: temperature-forced component hindcast (stacked) compared with NASA altimetry and reconstructions. The component sum closes the 2000–2025 budget to within 6%. **(f)** WAIS: IMBIE satellite observations (black) compared with the ISMIP6 ensemble median and interquartile range (green; [34]), our full scenario mixture (blue), the stable-WAIS scenario (purple), and IPCC AR6 total Antarctic projection (red).

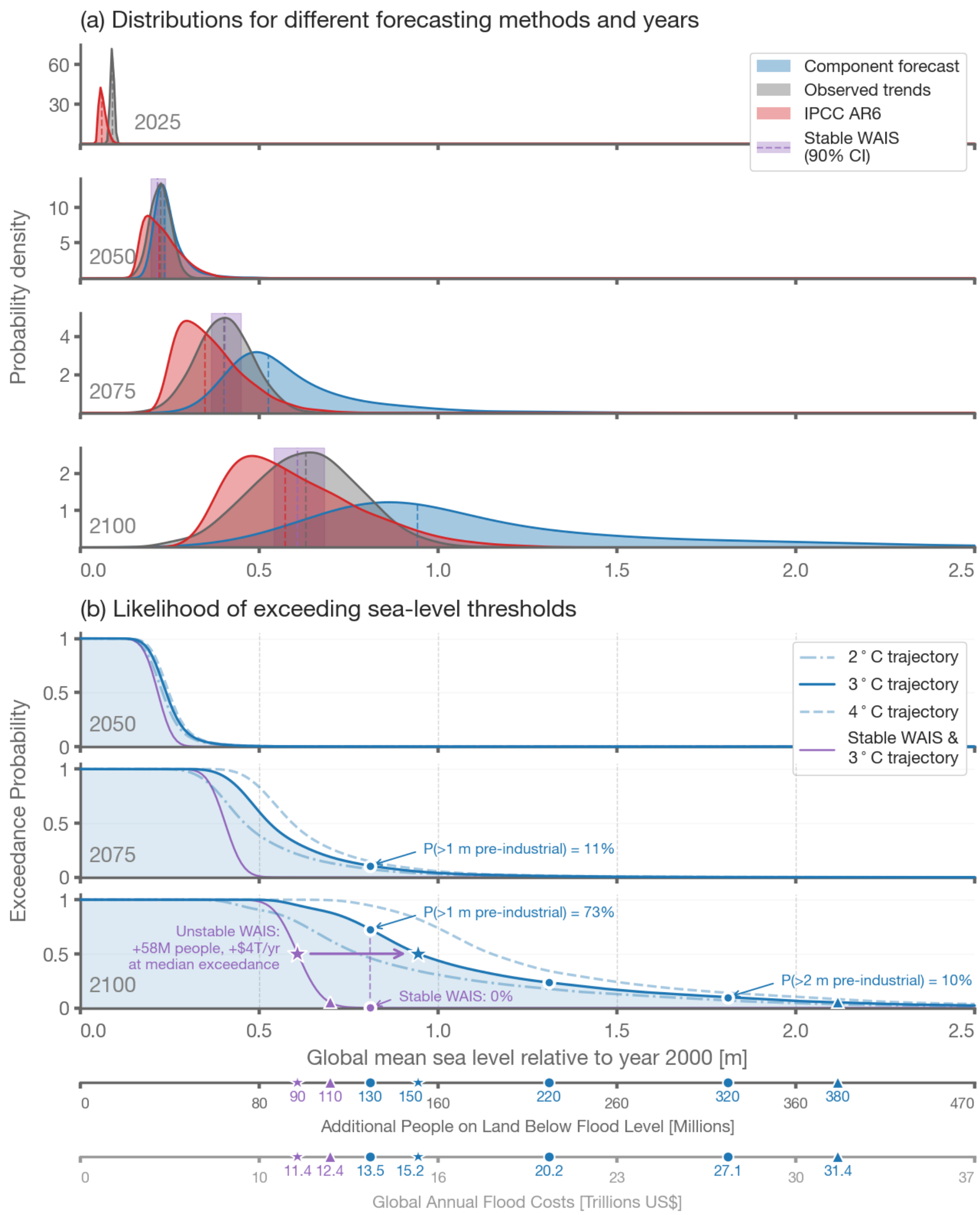

Figure 3: **Distribution and exceedance probability of global mean sea-level projections.** (a) Probability densities for the composite forecast of this study (blue), extrapolation of observations (grey; [2]), and IPCC AR6 projections (red; [12]). Vertical dashed lines indicate the respective medians. The purple band shows the 90% confidence interval for the case that WAIS does not experience a runaway acceleration, and sea level rise is due primarily to thermosteric, Greenland, and glaciers. We plot a purple band rather than distribution because the distribution is much tighter, and therefore taller, than the other distributions. (b) Probability of exceeding a given sea-level threshold for the 2°C (dash-dot), 3°C (solid), and 4°C (dashed) warming trajectories including instabilities in the West Antarctic Ice Sheet. Purple line shows the 3°C trajectory with a stable West Antarctic Ice Sheet. Lower axes show the corresponding number of additional people on land below flood level [7] and global annual flood costs [9], both assuming no additional coastal adaptation.